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Revisiting the evidence of the Arenal 1 site: Chronologies and human interactions in central southern Chile

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ABSTRACT

The Arenal 1 site in central southern Chile is renowned for its chicken DNA evidence suggesting pre-Columbian contacts with Polynesia. Debates around the chronology of the site and the lack of additional evidence supporting contact, gave origin to a new archaeological excavation focusing on refining the site's chronology and recovering further archaeological materials. The results appear to reconfirm a pre-Columbian date for this site; however, our findings did not produce any evidence of contact with Polynesia. Instead, clear evidence of South American interregional interaction was detected with the territory of Neuquén, Argentina.

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Introduction

Revisiting chronologies and archaeological contexts are essential to refining anthropological theories of cultural transmission and human interaction. New methods and approaches can provide additional evidence for understanding the developmental trajectories and cultural processes at play. In Polynesia, Transpacific contacts have been considered in the literature since at least the 1920s (e.g., Buck 1938; Dixon 1918; Emory 1942; Heyerdahl 1952; MacMillan-Brown 1924) and today there is no doubt that Polynesians could have reached the Americas (e.g., Ballard et al. 2005; Clarke et al. 2006; Green 2005; Hather and Kirch 1991; Jones et al. 2011; Kirch et al. 2017; Roullier et al. 2013; Scaglion 2005; Scaglion and Cordero 2011). However, there are still many questions around timing, regularity, effects, and directionality of such contacts. Chile has been argued to be a potential location of contact with Polynesia based on biological evidence and cultural parallels in material culture and linguistics. The Chilean evidence

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has included DNA evidence of pre-Columbian chicken remains recovered from the Arenal 1 site with links to remains found in Polynesia (Storey et al. 2007). It is important to note that these results have generated significant debate that is largely centered on contamination and chronological issues (Bryant 2014; Gongora et al. 2008; Storey et al. 2012; Storey and Matisoo-Smith 2014; Thomson et al. 2014).

Further evidence has included similarities in material culture including stone artifacts, marine technology, and linguistics (e.g., Jones 2011; Matisoo-Smith 2011; Matisoo-Smith and Ramirez 2010; Ramirez-Aliaga 1990, 2011). Various feasibility and simulation analyses utilizing environmental data have also shown that trips from Polynesia to South America and Chile were indeed possible (e.g., Finney 1994a, 1994b, 1994c; Fitzpatrick and Callaghan 2009; Goodwin et al. 2014; Irwin 1992, 2006). Certainly, Chile's extensive coastline of 6435 km and latitude of 17° South to Cape Horn at 56° would have provided a large target for intentional as well as accidental landings by Polynesian peoples.

Despite these efforts, the patchy nature of the evidence and the general lack of clear Polynesian archaeological contexts found in South America has limited further development in this line of research. To date no studies have produced conclusive evidence for and against Transpacific contacts in Chile prior to European arrival. A reevaluation of the Arenal 1 site is key to providing further insights into the Transpacific debate.

To reevaluate the chronology of the site and confirm its validity as evidence of pre-Columbian Transpacific contacts, a new archaeological investigation was initiated in October 2018. The main research objectives were to: (1) relocate the site; (2) reevaluate in detail the chronology of the various deposits through obtaining AMS dates of each stratigraphic layer presenting cultural materials; and (3) undertake specialist analyses on all cultural materials recovered, allowing evaluation of previous evidence regarding its chronology, material culture, faunal remains, and potential linkages with Polynesia (Buhring 2020). Important to this project was the creation of a multidisciplinary investigation team with specialists experienced in Pacific and central southern Chile archaeology to better understand the archaeological context of the site and materials recovered.

Environmental and geographical context

The Arenal 1 site is located on the northeastern border of the Yani sand dunes in the Arauco province, central southern region of Chile. Formally registered as PSAD56 623100 E/5863303N, this site is placed at about 1000 m south of the road to Quidico beach and about 30 m above sea level between the areas of Punta Lavapie and Lebu (Contreras et al. 2005), about 40 km from the city of Arauco, and 30 km from the city of Lebu, along the coastal route P-40 (Figure 1). The broader landscape is characterized by small fishing villages and farms and has been very isolated from the neighboring cities due to the lack of suitable roads. However, over the last decade, new motorway connections have allowed more regular contact with higher populated towns and cities. The local people are mainly of Mapuche descent and undertake farming, marine resources gathering, and diving as the main economic activities.

Geologically, the Arauco forearc basin covers an area of 8000 km^2 which extends 200 m below sea level and is the result of an uplifted part of the continental shelf

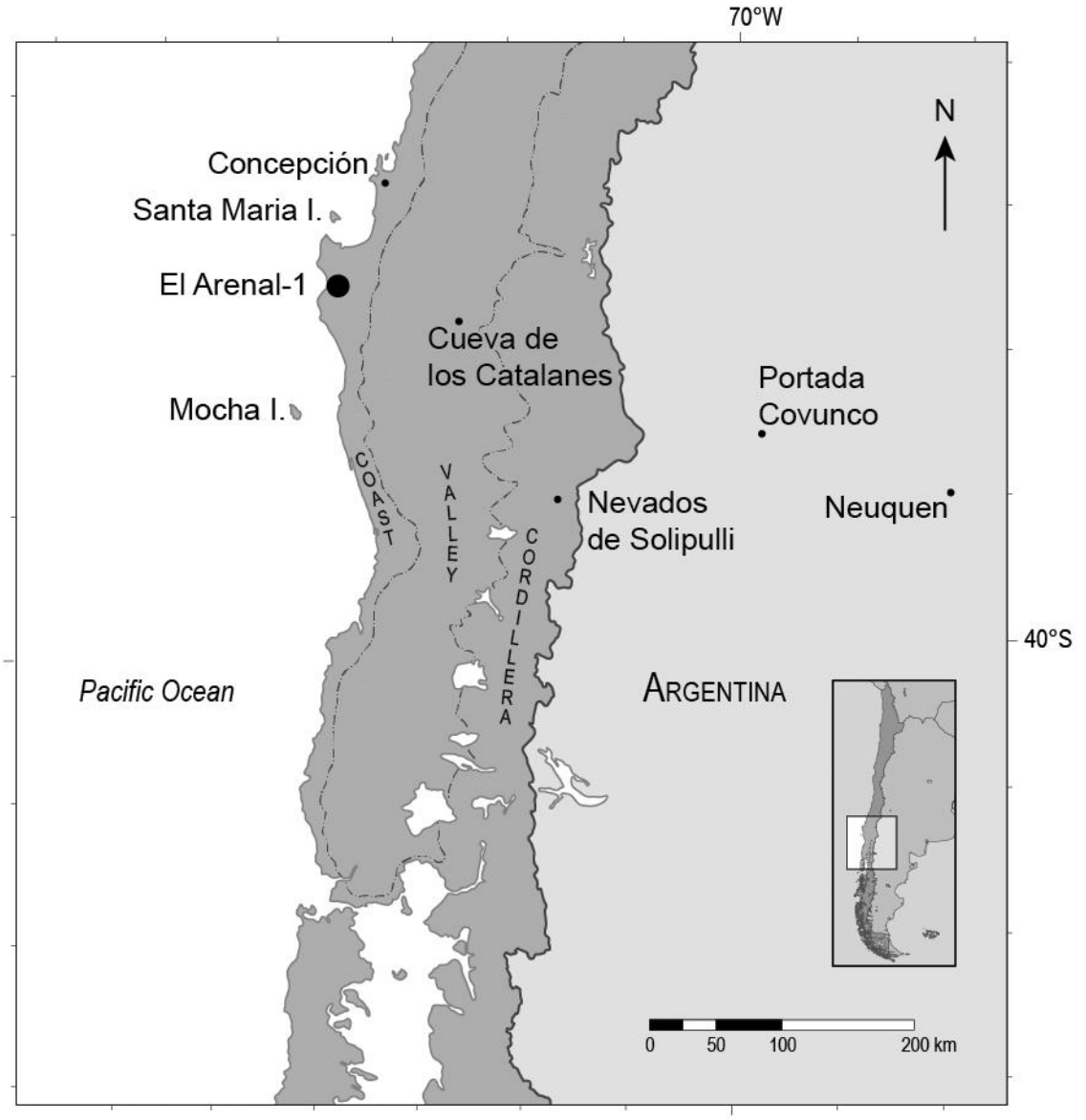


Figure 1. Location of the Arenal 1 site.

(González 1989; Mordojovich 1981). It is sedimentary in nature but is located between the Peru–Chile trench and a metamorphic accretionary complex intruded by the coastal batholith (Becerra et al. 2013). Morphologically, this area is characterized by a prevalence of accumulative geological forms which have given origin to a formation of transversal dune systems, wetlands, and floodplains associated with various streams and rivers that have contributed to local morphological erosion. The sand dunes of Yani are the result of the erosion of the coastal cordillera located to the east and sedimentary formations of marine origin surrounding the sand dunes. The sand dunes are currently active and continuously move inland in a northeast direction by prevailing

southwesterly winds (Parra 1999). The rate at which these sand dunes are advancing is currently unknown. This area is also affected by cyclical climatic changes which go from dry to cold and wet months. These changes give rise to seasonal streams and floodplains in the surrounding areas. As the Arenal 1 site is in a flood plain, during flooding episodes, organic and cultural material deposits of the site accumulate, creating optimal preservation conditions. Although the Arauco basin is constituted principally by sedimentary formations, the nearby rivers regularly transport metamorphic and volcanic rocks into this area. As such, all volcanic and metamorphic rock materials are considered as foreign to the site and have been brought in by rivers, streams, and/or by humans. This area is also regularly subjected to tectonic lifts. The last significant earthquake on February 27, 2010 produced uplifts of up to 2 m according to records taken at Punta Lavapie (Jaramillo et al. 2012). Furthermore, and due to the environmental characteristics surrounding the site, where deposits are made up of dry sand and wet materials, cultural deposits can also be shifted down and subjected to processes of compaction and dispersion. All of these are important considerations for archaeological investigations at the site.

Prehistory

The long human occupation in the central southern region of Chile was initiated in the Early Archaic Period with sites located in the Andean Cordillera dating to around 12,000 and 6,500 cal BP on the coast (Campbell and Quiroz 2015) although earlier dates have been set for the broader region (e.g., Perez et al. 2016). The Ceramic Period is subdivided into the Early Ceramic Period, defined by the Pitrén Cultural Complex (AD 350–1000), and the Late Ceramic Period, characterized by the El Vergel Cultural Complex (AD 1000–1550) (Adán and Mera 2011; Adán et al. 2016; Aldunate 2005; Bahamondes et al. 2014; Campbell 2011, 2014; Campbell and Quiroz 2015; Mera and Munita 2021; Quiroz and Sanchez 1997, 2005). The former has been defined by burials in flexed positions and ceramic stylistic indicators, including monochromatic designs, preference for darker colors, and emphasis on modeled and "negative technique" decorations. Pottery associated with the Pitrén Complex can be found within the area of the Bio Bio basin, on the northern side of Lake Llanquihue, Mocha Island extending to the east of the Andes into Provincia de Neuquén, Argentina (e.g., Adán and Mera 1997, 2011; Aldunate 1989; Hajduk et al. 2007; Menghin 1962; Ocampo et al. 2001; Quiroz and Sanchez 2005). Although broadly distributed, shared Pitrén elements are absent from early ceramic sites in the Gulf of Arauco area, and considerable cultural diversity has been recognized for this period (e.g., Adán and Mera 2011; Mera and Munita 2021; Palma 2016; Quiroz 2010; Quiroz and Sanchez 2005). The economy during this phase was mainly based on plant gathering and hunting, while there is also evidence of the introduction of new horticultural practices (Adán and Mera 2011; Campbell et al. 2018b). The El Vergel Cultural Complex is thought to have begun around AD 1000 and to have its origins in the Pitrén complex. The El Vergel complex has a broad distribution, covering the area of Angol and Toltén, including the islands, coast, and Andean valleys as far as Neuquén in Argentina (Navarro et al. 2011).

New cultural changes develop during this period, such as the construction of monumental architecture in specific areas (Campbell and Pfeiffer 2017; Dillehay 2007; Dillehay et al. 2007), emphasis on cultivated resources (although wild resources were still important) (Campbell et al. 2018b; Contreras et al. 2005; Massone et al. 2008; Roa 2016; Roa et al. 2018, 2021; Silva 2010, 2014), an intensification of a metalworking tradition (Campbell 2005), and an increase in long-distance exchange networks (e.g., Campbell et al. 2018a).

Changes in ceramic production are characterized by coarser and larger ceramics with red geometric motifs over white slip (Adán et al. 2005) but despite these changes, there is an overall continuity in vessel morphology from Pitrén.

Diverse mortuary practices also emerge during this period using mainly urns and burials with bodies in extended positions, and to a lesser extent, burials within canoes and possibly mounds (Adán et al. 2016; Bahamondes et al. 2006; Donoso 2002; Gordon 1978). The arrival of the Spanish in AD 1550 marks the end of the Late Ceramic Period and the beginning of the Historic Mapuche Period.

Archaeological context

The Arenal 1 site can be found between the northeastern border of the Yani sand dunes and the northwestern corner of a partially vegetated wetland south of the Quidico stream. It is also located to the west of a smaller stream and about 2 km east from the coast. The site was originally identified and recorded in 2002 (Quiroz 2003) as part of a comprehensive archaeological survey where over 400 archaeological sites were registered. Quiroz and his team identified a rich archaeology along the coastal section in the vicinity of the Arenal 1 site and 10 archaeological sites with varying levels of preservation were detected in the sand dune system itself. At this location, eight of these sites were interpreted as Archaic, one site was interpreted as historic, and the Arenal 1 site was defined as a habitation site and assigned to the El Vergel Period based on diagnostic surface materials and the results of a 1×1 m test pit excavation. During this preliminary investigation, materials recovered included ceramics, lithics made from basalt, andesite, and sandstone, shells from the rocky intertidal zone, fish, frogs, remains of camelids, and birds in deposits extending to 1 m in depth.

Later excavations by the same team (Contreras et al. 2005) confirmed the presence of ceramics found at the site as typical of the El Vergel ceramic complex. Lithics found included cores and debitage of basalt, andesite, quartz, and sandstone. Shell species found included those from the rocky intertidal zone, while faunal remains included birds, fish, camelids, canids, rodents, cetaceans, and amphibians. An earlier component was also identified as a possible pre-El Vergel occupation based on a thermoluminescence (TL) date of AD 700 ± 130 and the lack of El Vergel diagnostic red over white ceramics within the associated layer. During this excavation, a total of 50 remains of potential chicken (*Gallus gallus* sp.) were also recovered (Contreras et al. 2005), a number which increased to 83 remains during the subsequent 2007 excavations (Quiroz, et al. 2008). A subsample of these was sent to New Zealand for DNA analysis where chicken was identified with calibrated dates ranging from cal AD 1302 to 1425 (NZA 26115 622 ± 35 BP, 95% CI) (Storey et al. 2007).

The DNA results generated by Storey et al. (2007) indicated that these remains were identical to chicken bones from two prehistoric sites in the Pacific: Mele Havea in Tonga, dating to cal 795–423 BC (CAM 41521 2510 ± 50) (Burley et al. 2018: 6), and Fatu-ma-Futi in American Samoa, dating to cal AD 682–885 (WK-18525 1229 ± 30) (Addison and Asaua 2006: 100). Chickens are believed to have been introduced into the western Pacific region around 1000 BC as a result of the Lapita expansion, while the origins of modern chickens in central southern Chile were long thought to be associated with the Spanish conquest and subsequent occupation of the region. Storey's research was later challenged by Gongora et al. (2008) and Thomson et al. (2014) who raised issues concerning the reliability of dates and the contamination and validity of the genetic data. Further rebuttals and discussion of these issues can be found in Bryant (2014), Storey et al. (2011, 2012), Storey and Matisoo-Smith (2014).

Materials and methods

Archaeological investigations

Archaeological investigations were undertaken under archaeological permit Ord. Number: 1422 granted by the Consejo de Monumentos Nacionales (Chile's Heritage Agency) in 2018. This permit allowed three levels of assessment, which included auger testing (test cores), test pit investigation, and detailed archaeological excavation. Both auger testing and test pits assessments were carried out to locate optimum archaeological deposits, which were initially defined as meeting the following criteria: (1) the presence of abundant cultural material; (2) a well-preserved stratigraphy; (3) located in an area that could accommodate at least a 1 × 1 m excavation unit; and (4) as close as possible to the previous excavations as reported by Contreras et al. (2005) and Quiroz et al. (2006).

The coordinates reported originally for the site datum PSAD56 623100 E/5863303N were transformed into the currently used WGS84 resulting in coordinates 622862 E/5863012N. The excavation footprint included test cores, test pits, and excavation units (TP7, TP8, and TP9) as shown in Figure 2. A total of 34 cores of 10 cm in diameter were taken and each core sample was evaluated and recorded in detail. Subsequently, six test pits of 50 × 50 cm, or 50 × 100 cm were excavated. These test pits did not produce optimal deposits for further analysis, and some were also deemed unsafe due to unstable sand deposits and significant amounts of water. Test pits and materials were recorded, photographed, and evaluated. Three excavation units, TP7, TP8, and TP9 (Figure 2), were located at what appears to be the same location as the excavations undertaken previously by Quiroz based on photographic records, site descriptions, and information provided by the team, such as natural landforms, access, and fixed elements, including a fence in the hills to the north and immediately adjacent to the site and a wooden peg installed during the previous investigations. An open day and interactive session with the local school was held during excavations at the site which enabled us to share our preliminary findings with the local community. Subsequently, a series of laboratory analyses were undertaken in Chile, New Zealand, and England. These included faunal, malacological, macrobotanical, charcoal, ceramic, lithic, and DNA analyses.

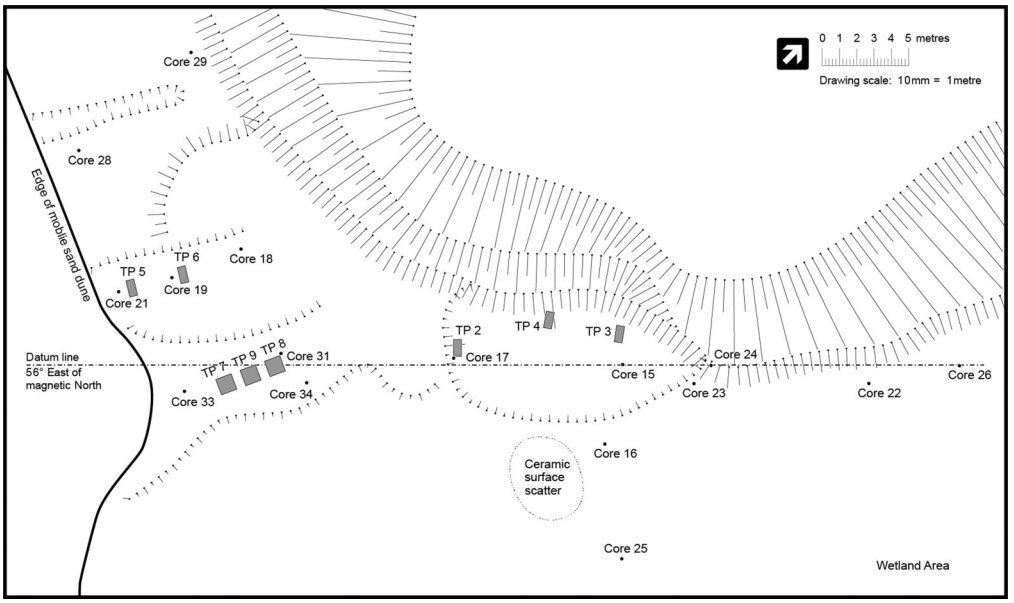


Figure 2. Plan showing main excavation area. Cores 1–14 were positioned south of the wetland area and therefore are not shown in this figure.

Stratigraphy

Excavations at the Arenal 1 site revealed a complex stratigraphy. A total of six layers were identified, four of which are associated with cultural episodes (Layers 2, 3, 4, and 5) (Figure 3). The cultural layers are characterized by a significant amount of organic and inorganic materials with excellent preservation. The archaeological excavation units showed an overall continuity in terms of stratigraphy and what appears to be displacement of deposits possibly due to earthquake activity.

Located directly under the grass cover, Layer 1 comprises unconsolidated fine-grain sands. These are yellowish brown and composed predominately of quartz and some feldspars, ferromagnesium minerals, very fine-grained metamorphic lithic fragments, and some bioclasts (shell). Generally, the sand grains are angular, indicating low-level transport and are likely local in origin. It is possible that this layer originated in the nearby sand dunes, which also show the same characteristics on the surface. There is also some mud content indicative of flooding events. There is no cultural material in this layer.

Layer 2 is a dark brown, very fine sand which is principally quartz, although in smaller quantities compared to Layer 1. There is a higher amount of ferromagnesian and lithic material of metamorphic and volcanic origin based on the observed disposition and color of its minerals. This layer also includes silty materials. This matrix is interpreted as being formed by sands derived from the adjacent dunes, but with a contribution of volcanic materials from the nearby river during past flood events. In contrast to Layer 1, Layer 2 contains cultural features and materials, such as ceramics, bone, shell, and charcoal fragments, among other site constituents.

Layer 3 is a dark very fine sand with a significant increase in the amount of mud and organic material. Its composition shows a decrease in the percentage of quartz and

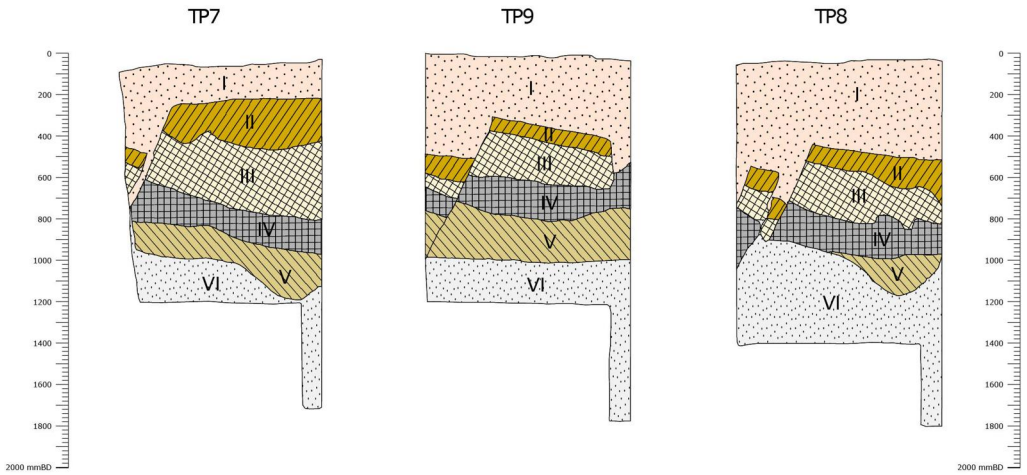


Figure 3. Stratigraphic plan of main excavation units at the Arenal 1 site (south facing walls).

ferromagnesian minerals. Lithic and feldspar grains are present, but in very low amounts. These sands appear to be a mixture of dune and alluvial materials with clear anthropic factors which have resulted in a higher quantity of charcoal present throughout the matrix. Cultural material increases significantly and includes shell, bone, lithic, charcoal, and ceramics. Layer 4 is similar to the above, with a high content of silt and clay but significantly less organic material. Some possible cultural features were observed, potentially suggesting drains or structural imprints. Cultural materials recovered include charcoal, ceramics, bone, and shell.

Layer 5 is similar to Layer 4 but contains brown mottled sand. Cultural material recovered from this layer includes charcoal, shell, lithics, ceramics, and bone. Layer 6 is a clean light-yellow sand with no cultural material, although a feature containing archaeological materials from Layer 5 was cut into this layer in TP7. These findings are generally consistent with previous stratigraphic descriptions (e.g., Contreras et al. 2005).

Results

Radiocarbon dates

Samples for AMS were taken from Layers 2, 3, and 5 in unit TP7 to get chronological information for the overall stratigraphic sequence. A careful selection of charcoal samples was undertaken at a transitional facility (Biomolecular lab) at the University of Auckland. To improve the accuracy of dates, only short-lived species were selected for AMS analysis. Samples were neither cleaned nor treated prior to submission. A total of four samples were sent to the Radiocarbon Dating Laboratory at the University of Waikato.

Overall, our new dates (Table 1) are consistent with those reported by Storey et al. (2007) and Storey (2009), and therefore, it is possible to assume a human occupation within the period from cal AD 1300 to 1640. Notably, the TL date of AD 700 ± 130 given by Contreras et al. (2005) for their earliest Layer C is inconsistent with our date

Table 1. Radiocarbon and TL dates available for the Arenal 1 site.

Stratigraphic information	Lab code	CRA and TL 1 SD results	Cal date 95% confidence	Sample information	Reference
Layer 5 (context 51)	WK50049	614 ± 21	AD 1321–1417	3 mm diameter stem	
Layer 5 (context 50)	WK50050	412 ± 21	AD 1453–1623	Corn	
Layer 3 (context 32)	WK50048	353 ± 21	AD 1500–1643	Monocot tissue	
Layer 2 (context 21)	WK50051	428 ± 21	AD 1450–1619	3 mm diameter stem	
?	NZA26115	622 ± 35	AD 1302–1426	Chicken bone	(Storey et al. 2007)
?	NZA28271	510 ± 30	AD 1408–1459	Chicken bone	(Storey 2009)
?	NZA28272	503 ± 30	AD 1408–1463	Chicken Bone	(Storey 2009)
Layer A	Unknown	1350 ± 65 AD	Thermoluminescence	Pottery (El Vergel complex)	(Contreras et al. 2005)
Layer B	Unknown	1390 ± 55 AD	Thermoluminescence	Pottery (El Vergel complex)	(Contreras et al. 2005)
Layer C	Unknown	700 ± 130 AD	Thermoluminescence	Pottery (Unknown cultural complex)	(Contreras et al. 2005)

Note: Calibrated with OxCal 4.4.4 (Bronk Ramsey 2021) using SHCal 20 atmospheric curve (Hogg et al. 2020).

for the lowest cultural layer (Layer 5) which resulted in cal AD 1319–1416 (WK50049, 95% CI) and the other available dates for the site.

Ceramics from the Arenal 1 site share similar characteristics with other El Vergel archaeological sites (e.g., Campbell 2011; Contreras et al. 2005; Gajardo 2017, 2019; López 2014, 2016). Typical diagnostic El Vergel Period ceramics were found throughout the entire deposit during our excavations at the site confirming its chronology.

Faunal analysis

The analysis of faunal remains consisted of two parts: DNA identification at the University of Otago (Collins and Matisso-Smith 2019) and a complete zooarchaeological taxa identification of 109 samples which was carried out in Chile using local reference collections. The first stage involved the visual identification of untreated and/or uncontaminated remains to select potential chicken samples for DNA analysis. Visual identification and selection of bird samples was first conducted at the transitional facility (Biomolecular lab) in the Department of Anthropology at the University of Auckland utilizing Quiroz et al. (2008) morphometric values for the reported chicken samples. Interestingly, in that study, by comparing the Arenal 1 site humerus, coracoid, tibiotarsus, and tarsometatarsus bones with reference collections from other chicken species such as Araucana, Sumatra, and Passion fowl, the authors concluded that the Arenal 1 chickens were more similar in size to the passion fowl but had shorter legs and longer wings, probably indicating better flying capacity (Quiroz et al. 2008, 104).

From the bird samples identified in our investigations, three specimens from the Arenal 1 samples were deemed as potentially chicken and were selected as candidates for DNA analysis at University of Otago. None of the samples analyzed were determined to be chicken bones, but instead showed affiliation with tinamou (Tinamidae) (Collins and Matisso-Smith 2019). Tinamidae are native to North, Central, and South America, and among their phenotypical characteristics is their ability to fly in short bursts. The second stage of faunal analysis was undertaken in Chile using local reference collections where 17 taxa were identified. While this is a relatively small sample, a high diversity of taxa was identified with fish and birds constituting the main assemblages.

Species included mammals such as whales (Cetacea), camelids (Camelidae), coipo (*Myocastor coypus*), rodents (Sigmodontinae), small carnivore, reptiles (e.g., *Liolaemus* sp.), frogs (*Calyptocephalella gayi*), several species of fish, and birds, including rails (Rallidae), cormorants (*Phalacrocorax sp.*), seagulls (Laridae), Passeriformes, and tinamou (*nothoprocta perdicaria*). Butchering signs were observed on Camelidae bones. We also found two undetermined polished artifacts, one made from whale bone and the other of undetermined mammal bone. Additionally, a thin, circular-shaped polished artifact made of undetermined mammal bone was also recovered. During our investigation, we were unable to locate chicken remains as per previous excavations; however, we did discover tinamou (Tinamidae) remains not previously reported at this site (refer to supplementary information). A subsequent morphological reassessment of 66 samples previously identified as chicken by Quiroz et al. (2008) provided no evidence of chickens but were in fact tinamou. Figure 4 illustrates morphological indicators utilized in our taxonomical analyses.

Malacological analysis

A total of 965 mollusk remains were recovered from our excavations. Analysis focused on taxa identification, NISP and MNI estimation, determining taxa distribution, taphonomic patterns, as well as identifying use and consumption patterns. The taxonomic diversity found at the site is considered relatively low with a clear preference for gastropods. This is interesting as the intertidal environment nearby is a habitat rich in bivalve species but not gastropods, which are usually found in hard rocky areas such as those located approximately 2 km north and south of the site. Gastropods are mainly represented by the Chilean abalone (*C. concholepas*), limpets (*Fissurella* sp.), and sea black snails (*Tegula atra*) as well as other sea snails (*Acanthina monodon*). Bivalves include taca clams (*Protothaca thaca*) and macha clams (*Mesodesma donacium*). Marine crabs were also found but in very low abundance. The habitats most represented in the sample were intertidal and rocky subtidal environments, although species from all other habitats were also represented. Taphonomic evidence of burning, carbonization, and bleaching are indicative of typical processing and consumption of marine resources. The high degree of fragmentation observed also suggests trampling resulting from repeated use of the site. Fragmentation could also relate to the fragile characteristics of shell remains, particularly snails (*Tegula atra*) and mussel choro zapato (*Chroromytilus chorus*). The earliest occupation (Layer 5) shows a more significant variety of marine resources. The relative abundance of species from the rocky intertidal and their easy accessibility at low tide, suggest that these were easily harvested. These results indicate consumption of a variety of marine resources that were locally available along sandy and rocky intertidal zones commonly found along the Chilean coast.

Macrobotanical analysis

A total of 22 sediment samples were recovered from two profile columns in Units TP7 and TP8 once excavation was completed. Macrobotanical remains were recovered through bucket flotation. Recovery rates of samples were estimated by controlled

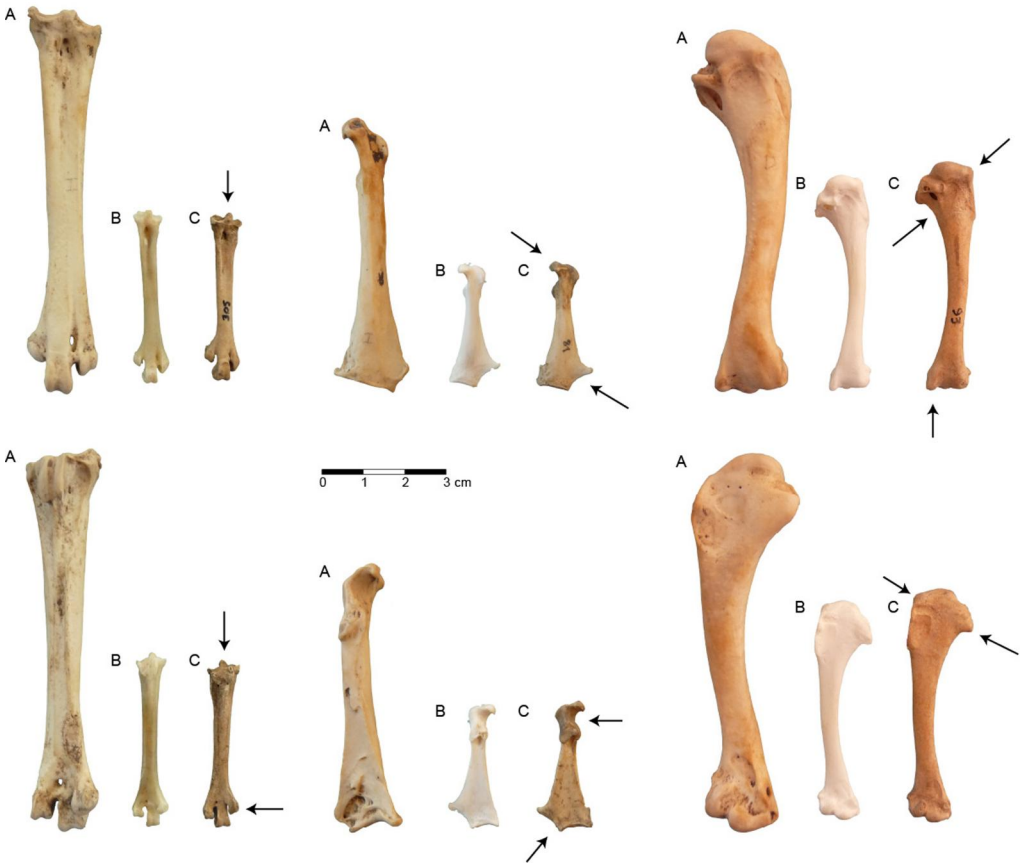


Figure 4. Bird comparison examples. Top row: anterior view. Bottom row: posterior view: (A) Domestic chicken (reference collection); (b) tinamou (reference collection); (c) the Arenal 1 reported chicken (Quiroz et al. 2008). Bone types and examples of diagnostic anatomical differentiators in arrows (from left to right): tarsometatarsus: eminentia intercotylaris, intertrochlearis medialis; coracoid: procesus acrocoracoideus, procesus lateralis; humerus: tuberculum dorsale, crista bicipitalis and processus flexorius.

experimental tests. The species represented at the site include native species such as *madi* (a species of native tarweed, *Madia sativa*), *maqui* berry (*Aristotelia chilensis*), *chaura* shrubs (cf. *Gaultheria* sp.), wild strawberry (cf. *Fragaria chiloensis*), wild raspberry (*Rubus* sp.), madder family species (cf. Rubiaceae), and grasses (*Bromus* sp. and various undetermined Gramineae). Among the introduced species are quinoa (*Chenopodium quinoa* Willd.) and maize (*Zea mays* L.). Quinoa, maize, wild raspberry, and undetermined Gramineae were the most ubiquitous taxa identified at the site. These taxa are well distributed in the stratigraphic sequence demonstrating a clear consumption of both cultivated and wild taxa. Current dates for earliest maize in the region are estimated to be around AD 900–1000 while quinoa has been set to cal AD 600 (Campbell et al. 2018; Roa et al. 2018). A maize seed from the early Layer 5, Context 50, produced a date ranging from cal AD 1453 to 1623 (WK50050 412 ± 21 BP, 95% CI) suggesting these crops were introduced late to the Arenal 1 site. This study was able to identify seven new taxa for the site, which includes goosefoots (*Chenopodium* sp.), *madi*

(*M. sativa*), *maqui* (*A. Chilensis),* brome (*Bromus* sp.), wild strawberry (cf. *F. chiloensis*), *chaura* (cf. *Gaultheria* sp.), and madder family species (cf. Rubiaceae). No macrobotanical remains of sweet potato were identified.

Based on the results of our macrobotanical analysis, the Arenal 1 site shows similarities to what is found at other El Vergel archaeological contexts, including Santa Maria Island (Massone et al. 2012), Mocha Island (Roa et al. 2016; Roa et al. 2021; Rojas and Cardemil 1995; Sánchez et al. 2004), Cueva de los Catalanes (Roa et al. 2018), and Puren and Lumaco Valley (Dillehay et al. 2007; Silva 2014) whereby a mix of wild species were gathered and introduced species such as quinoa and maize were also cultivated.

Charcoal analysis

Charcoal identification was undertaken at the Institute of Archaeology, University College London. The aim of this analysis was to generate additional environmental information for the site as no charcoal data were previously examined. A total of 1537 charcoal fragments were analyzed and 20 different native woody taxa were identified. The taxonomic identification was done by direct comparison using a reference collection of charred modern specimens (more than 40 woody taxa) and in both published and unpublished literature on South American woody species (e.g., Rancusi et al. 1987; Torricelli 1937; Tortorelli 2000; Wagemann 1949).

In ecological terms, these results show the presence of evergreen taxa such as *olivillo* (*Aextoxicon punctatum*), *Amomyrtus* sp., Cunoniaceae, Monimiaceae, Myrtaceae, *lingue* (*Persea lingue*), and sclerophyllous taxa such as *peumo* (*Cryptocarya alba*), Lauraceae, *litre* (*Lithrea caustica*), *boldo* (*Peumus boldus*), *quillay* (*Quillaja saponaria*). Smaller trees and bushes were also identified including *maqui* berry (*Aristotelia chilensis*), *matico* (*Buddleja globose*), *Maytenus* sp., Liliopsida, *Rhaphithamnus spinosus*, *Ribes* sp.

Initially, these results indicate that the surroundings of the Arenal 1 site were characterized by an ecotone of evergreen coastal forest and Mediterranean sclerophyllous vegetation. The predominance of the genus *Amomyrtus* could indicate that these taxa were available during the site's occupation. The *Amomyrtus* genus has only two species: *Amomyrtus luma* and *Amomyrtus meli*, commonly known as "luma" and "meli." Both are evergreen species, and in terms of plant succession, they have been classified as intermediate species because they can tolerate both sunlight and shaded areas under close canopy forest or in more open landscapes. In terms of potential use, both species of the *Amomyrtus* genus have resistant hard wood used to create tool handles (Hoffmann 2005). *Amomyrtus luma* has edible berries (Corderoet al. 2017) and, although *Amomyrtus meli* also has sweet and fragrant berries, its consumption has not been conclusively reported (García and Ormazabal 2008).

Other taxa identified include species with known medicinal use (e.g., *Peumus boldus* and *Buddleja globosa*), as well as taxa with edible fruits (*Cryptocarya alba*, *Ribes* sp., and *Aristotelia chilensis*) (Mösbach 1992). In general, all the identified taxa have good firewood potential, and even small woody plants, such as *Rhaphithamnus spinosus* and some *Liliopsida* or monocots plants, have been described as suitable for starting fires (Gumucio 1999).

Ceramic analysis

Ceramic analysis involved morphological, stylistic, and functional characterization of 1435 ceramic fragments (133 decorated, 941 non-decorated, and 361 undetermined). The formal analysis of ceramic fragments suggests a preference in use of undecorated vessels with smoothed exterior walls with varying levels of thickness, ranging from medium to very thick walls. Typical El Vergel decorated fragments (red over white painted decoration) were also detected throughout the entire deposit. It was possible to identify bowls or *pucos* of various diameters, jugs, and necked pots with various rim types. Some of the very thick fragments may indicate the presence of domestic (non-funerary) urns reported previously for the site (Contreras et al. 2005) and characterized by the presence of soot adhered to surfaces. A small perforation was also identified in an urn fragment which indicates it was repaired to extend its use life. Soot was found on jugs and pots fragments which demonstrates their exposure to fire, suggesting a domestic El Vergel context. These characteristics are also similar to those observed at other El Vergel archaeological contexts (e.g., Bahamondes 2009; Bahamondes et al. 2014; Cortés et al. 2016; Donoso 2017; Gajardo 2017; López 2014, 2016; Reyes 2010) within central southern Chile.

Lithic analysis

Lithic analysis included visual rock identification and technological and geochemical characterization at the University of Auckland. The complete assemblage recovered from the excavation totaled 109 specimens. Various artifacts representing only primary and secondary debitage typologies—mainly in quartz and fine-grained igneous rocks—were recovered, indicating an expedient technology. Two cores were also identified in the sample, one from fine-grained igneous rock and another made with sandstone, both of which showed unidirectional flaking. Our findings indicate that lithic processing activities at the site were minimal and focused on the manufacture of scrapers/cutting artifacts of local pebbles. With the exception of a hammerstone, most instruments appear to come from small sized rocks probably collected in the vicinity of the site, some of them showing signs of erosion, potentially reflecting fluvial transport.

A tiger striped obsidian arrowhead fragment (Sample 187) found at the site was geochemically analyzed to confirm its source. Sample 187 was compared to Patagonian source data reported in Stern (2018). Of the 22 sources included in Stern's tables, Sample 187 is the closest geochemical match to Portada Covunco, in north-central Patagonia, Argentina (Figure 5), and also matches the trace element source and artifact data reported in Campbell et al. (2017) as shown in Table 2. This type of obsidian was widely distributed and has been identified as far west as Mocha Island (Campbell et al. 2017). Notably, Sample 187 possesses a black and red tiger striped color which is also considered to be distinctive of Portada Covunco obsidians. Overall, these data suggest that the Arenal 1 site was also part of an interregional trade network that involved acquiring obsidian raw materials and/or finished artifacts from the other side of the Andes.

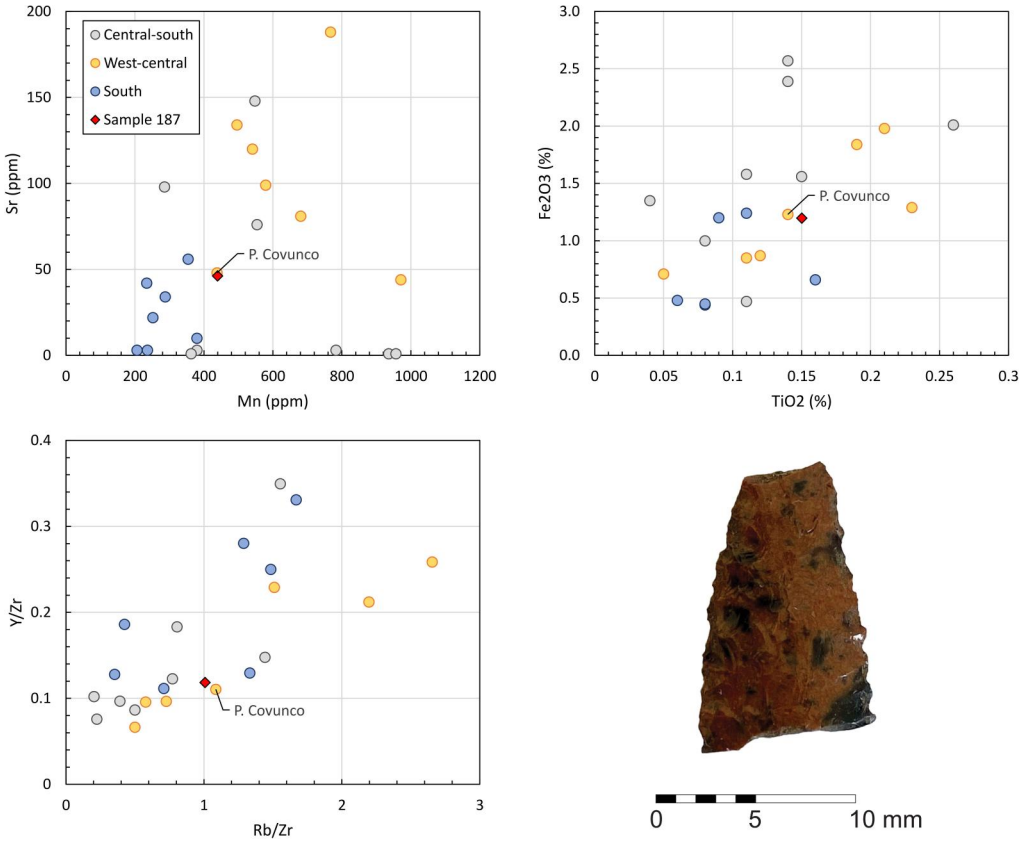


Figure 5. Scatterplots of Mn vs Sr, TiO_2 vs Fe_2O_3, and Rb/Zr vs Y/Zr comparing Sample 187 to the source averages reported by Stern (2018). Image of the tiger striped obsidian arrowhead fragment Sample 187.

Table 2. Comparison of Sample 187 to Portada Covunco source data reported by Campbell et al. (2017) and Stern (2018), and the Mocha Island artifact attributed to Portada Covunco (Sample 29) by Campbell et al. (2017).

		P. Covunco (Campbell et al.2017)	Sample 29 (Campbell et al. 2017)	P. Covunco (Stern 2018)	Sample 187 Arenal 1
SiO_2	%			75.52	74.05
TiO_2	%			0.14	0.14
Fe_2O_3	%			1.23	1.23
MnO	%			0.05	0.06
CaO	%			0.54	0.62
K_2O	%			4.71	4.54
Ti	ppm		1143	982	839
Mn	ppm		405	438	437
Rb	ppm	177	171	177	171
Sr	ppm	46	48	48	46
Y	ppm	18	19	18	20
Zr	ppm	157	158	163	169
Nb	ppm		28	29	30
Th	ppm		25.1	26.8	28.9

Discussion

Radiocarbon dates and the material culture recovered from the Arenal 1 site are clearly suggestive of an El Vergel occupation which probably began sometime around AD 1300 and appears to have lasted until prior to Spanish arrival. The archaeological assemblages recovered from the site confirm that occupants here were extremely knowledgeable about their environment as demonstrated by their consumption of a variety of locally available terrestrial and marine species. Macrobotanical evidence also indicates the consumption of quinoa and maize that had already been introduced into the region along with consumption of several local plant species. Traditional knowledge of the occupants of the Arenal 1 site also included the production of red over white ceramics, characteristic of El Vergel sites. Based on the material evidence recovered to date, we concur with previous interpretations of the site as a habitation/domestic context (Contreras et al. 2005). During our excavations, we also recovered fragments of quincha ($n = 203$), a mixture of clay and vegetal fibers, which has been associated with the construction of walls in domestic structures (Boiser 2013). While the lithic assemblage is small, it does show the exploitation of primarily local materials, some of which were likely recovered from nearby streams with some exotic materials also present.

The chronology of the Arenal 1 site points to a late El Vergel occupation beginning around cal AD 1300 and lasting through to cal AD 1640, which is consistent with previous radiocarbon dates obtained for the site. Although these dates appear to extend into the Historic Period, no European material culture has yet been found. The El Vergel red over white ceramics that are present through the entire sequence confirms this temporality. Thus, the evidence available to date for this site strongly suggests a human occupation consistent with the El Vergel Period.

Transpacific interactions

Based on these results, there is no evidence at the Arenal 1 site to suggest an external non-American influence, except for the previously reported chicken bone DNA, which our 2018 excavations and subsequent analyses failed to replicate. Instead, our study detected tinamou remains (*Nothoprocta perdicaria*) which were confirmed by both visual zooarchaeological identification and through DNA profiling. Tinamou was not reported in any of the previous excavations at the site, but its consumption has been reported at other El Vergel sites (e.g., Quiroz and Martinez 2010). Spanish accounts have also described regular tinamou consumption by Mapuche people (e.g., Núñez de Pineda y Bascuñán 1673). Furthermore, our morphological reassessment of samples previously reported as chicken (Quiroz et al. 2008) have reclassified them as tinamou. This makes previous DNA data appear inconclusive (Storey et al. 2007). Unfortunately, no remaining bone samples from that study were available to be reanalyzed.

Sweet potato still constitutes the primary evidence for a South American–Polynesia connection. Sweet potato remains recovered from archaeological deposits in Mangaia (Cook Islands) (Hather and Kirch 1991) dating to cal AD 991–1261 (Beta-32826; 980 ± 70 BP) have often been considered as evidence for sweet potato's initial introduction into the Pacific. Currently, the scenario sets its arrival to Mangaia, sometime after AD 1400 and not during initial colonization (Kirch et al. 2017; Niespolo et al. 2019). Similarly, the earliest dates in New Zealand and Hawai'i are also around AD 1400 (Anderson and Petchey 2020; Coil and Kirch 2005;

Ladefoged, Graves, and Coil 2005) while Rapa Nui and the Marquesas may have earlier dates ca. AD 1200–1400 (Allen and Ussher 2013; Horrocks and Wozniak 2008; Tromp and Dudgeon 2015). Various lines of evidence, ranging from the sweet potato to linguistics, genetics, computer simulations, and ethnographic accounts, appear to support an Ecuador/northern Peru–Marquesas connection (e.g., Ioannidis et al. 2020; Roullier et al. 2013; Scaglion 2005; Scaglion and Cordero 2011) although some theories of oceanic drift have also been explored (e.g., Montenegro, Avis, and Weaver 2008; Muñoz-Rodríguez et al. 2018; Temmen et al. 2022). Early ethnographic accounts (e.g., De la Vega 2004; Sarmiento de Gamboa 1572, in Ortega 2018) also suggest Ecuador and northern Peru as key areas for further investigation and computer simulations suggest that further evidence may be found along the northwestern South American coast (Fitzpatrick and Callaghan 2009). As such, these may be fruitful geographic areas to focus any future Transpacific research.

Interregional interactions

The presence of a tiger striped obsidian projectile point fragment at this site is significant. This arrowhead fragment (Sample 187), geochemically linked to the west of Neuquén province in Argentina, demonstrates that the Arenal 1 people were also part of an interregional interaction network which has been detected in other El Vergel sites such as on Mocha Island (e.g., Campbell et al. 2018; Campbell et al. 2017; Peñaloza et al. 2019). Tiger striped obsidians are rare but do appear at archaeological sites on both the Pacific and Atlantic coasts of South America (Stern 2018). Previous studies at the Arenal 1 site recovered a stemmed arrowhead made with black obsidian (Contreras et al. 2005) which has yet to be geochemically linked to its source. As previously found on Mocha Island, there were various types of obsidians brought into this region, including the tiger striped, black, and gray obsidians with origins in distant areas, near the Andes cordillera such as Portada Covunco and Nevados de Solipulli (Campbell et al. 2018). Interregional interaction is also suggested by compositional ceramic evidence between El Vergel groups within this region and nearby islands (Gajardo 2019). Further geochemical and compositional studies would be extremely useful in continuing to develop our understanding of interregional human interaction networks during this period.

As shown in our investigation of the Arenal 1 site, archaeological research that takes advantage of multidisciplinary approaches for examining a wide range of materials is key to resolving questions regarding complex cultural behaviors and interactions that occur through time and across space. In addition, the reexamination of archaeological materials such as bird bones at the Arenal 1 site which have been used to argue for Polynesian connections to South America have important implications for determining when, how, or even if long-distance dispersions were in fact occurring in this region.

Supplemental Material

Supplemental Table S1. Faunal taxonomic diversity.
Supplemental Table S2. Malacological taxonomic diversity.
Supplemental Table S3. Macrobotanical taxa diversity.
Supplemental Table S4. Charcoal taxonomic identification.
Supplemental Table S5. Ceramic analysis.
Supplemental Table S6. Lithic analysis.

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